

Laboratory Equipment Booking & Maintenance System

Presentation III – User Interface Demonstration

Mallavarapu Suchith • 25WU0102144

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STATUS

Connected to MySQL

UI Overview

A simple interface for interacting with the existing lab_equipment database.

Dashboard

Live summary of equipment, bookings

Equipment

View, add and delete equipments

Bookings

View booking records and create new b

Consumables

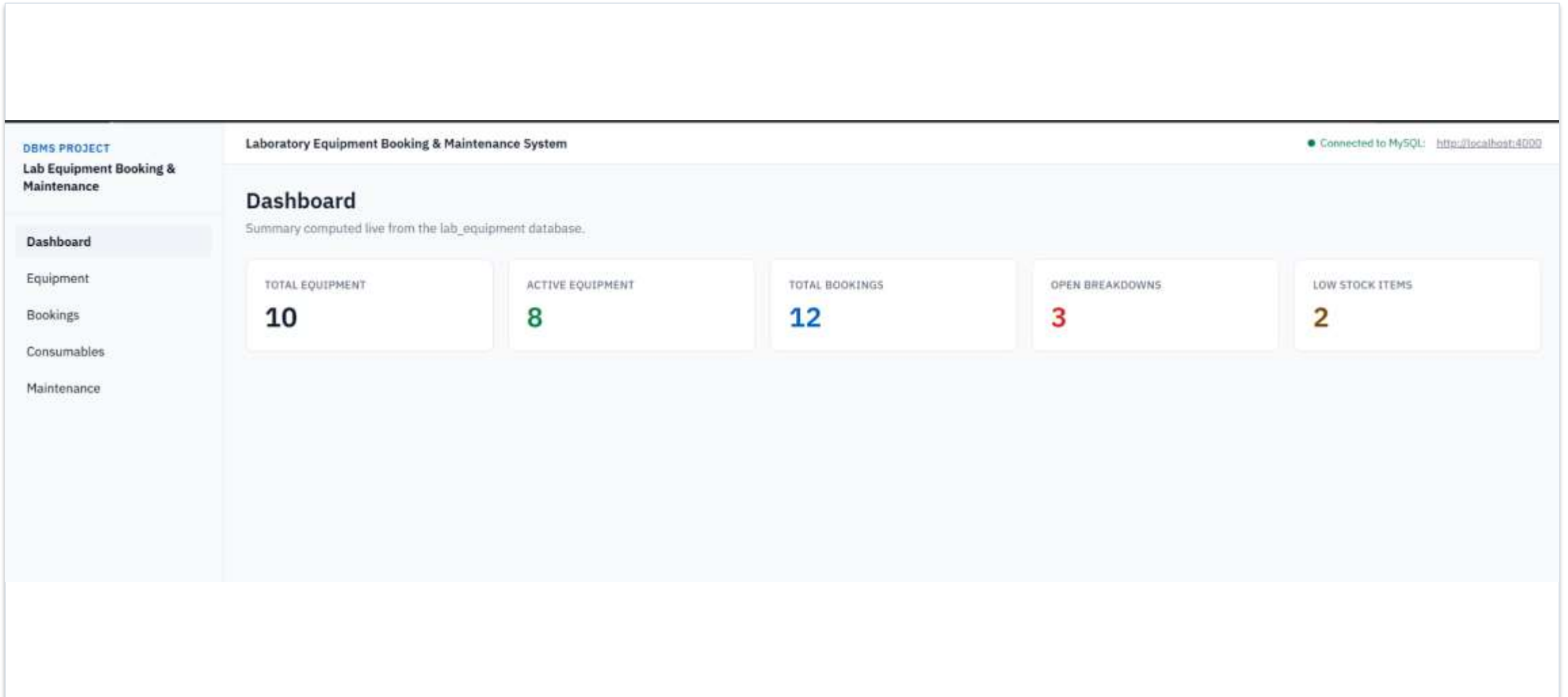
Monitor quantities and identify l

Maintenance

View maintenance jobs, breakdowns and vendors.

Dashboard

Live summary computed from the database.



Equipment Records

Equipment data displayed with category and laboratory information.

DBMS PROJECT Lab Equipment Booking & Maintenance Dashboard Equipment Bookings Consumables Maintenance	Laboratory Equipment Booking & Maintenance System						Connected to MySQL: http://localhost:4000
	Equipment Equipment > Equipment_Category > Laboratory + Add Equipment						
	EQUIPMENT ID	EQUIPMENT NAME	CATEGORY	LABORATORY	STATUS	ACTIONS	
	101	DSO-1000	Oscilloscope	Physics Lab	Active	Delete	
	102	DSO-2000	Oscilloscope	Physics Lab	Active	Delete	
	103	Digital Multimeter	Multimeter	Electronics Lab	Active	Delete	
	104	Microscope X1	Microscope	Electronics Lab	Maintenance	Delete	
	105	3D Printer Pro	3D Printer	Computer Lab	Active	Delete	
	106	Centrifuge C200	Centrifuge	Chemistry Lab	Active	Delete	
	107	SolderPro X2	Soldering Station	Electronics Lab	Active	Delete	
	108	Function Generator FG1	Function Generator	Physics Lab	Active	Delete	
	109	Centrifuge C300	Centrifuge	Chemistry Lab	Broken	Delete	
	110	solderpero x3	Soldering Station	Electronics Lab	Active	Delete	

Insertion of a Record

Demonstrate a real INSERT through the Equipment interface.

01

Open Add Equipment

Click "+ Add Equipment" on the Equipment page.

02

Enter details

Provide equipment ID, name, category, laboratory

03

Submit

Save the record through the UI.

04

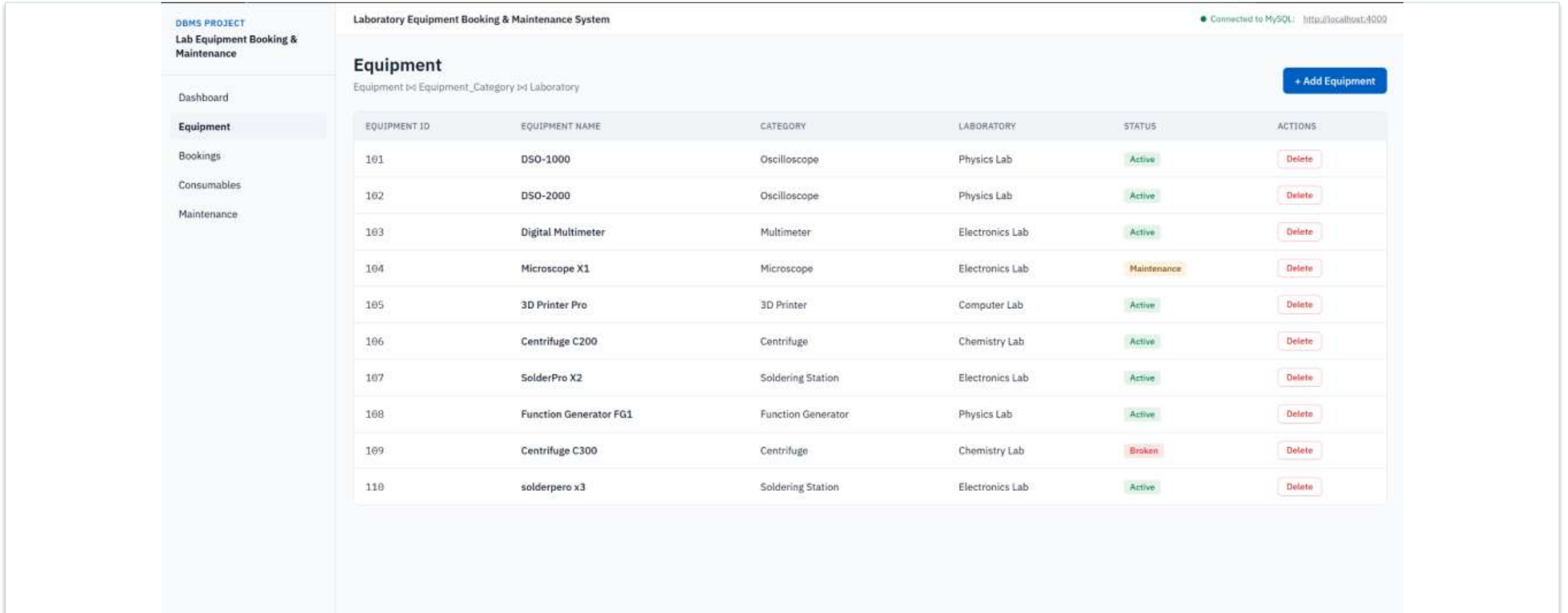
Verify

Refresh the list and verify the new row in MySQL.

Live demo: BEFORE → ADD → UI UPDATE → MySQL verification

Viewing Records

The UI retrieves and displays records from MySQL.



The screenshot displays the 'Equipment' management interface of the 'Laboratory Equipment Booking & Maintenance System'. The interface includes a sidebar with navigation links: Dashboard, Equipment (selected), Bookings, Consumables, and Maintenance. The main content area shows a table of equipment records with columns for Equipment ID, Equipment Name, Category, Laboratory, Status, and Actions. A '+ Add Equipment' button is located in the top right corner. The table contains 10 rows of data, with the 9th row (ID 109) marked as 'Broken'.

EQUIPMENT ID	EQUIPMENT NAME	CATEGORY	LABORATORY	STATUS	ACTIONS
101	DSO-1000	Oscilloscope	Physics Lab	Active	Delete
102	DSO-2000	Oscilloscope	Physics Lab	Active	Delete
103	Digital Multimeter	Multimeter	Electronics Lab	Active	Delete
104	Microscope X1	Microscope	Electronics Lab	Maintenance	Delete
105	3D Printer Pro	3D Printer	Computer Lab	Active	Delete
106	Centrifuge C200	Centrifuge	Chemistry Lab	Active	Delete
107	SolderPro X2	Soldering Station	Electronics Lab	Active	Delete
108	Function Generator FG1	Function Generator	Physics Lab	Active	Delete
109	Centrifuge C300	Centrifuge	Chemistry Lab	Broken	Delete
110	solderpero x3	Soldering Station	Electronics Lab	Active	Delete

- Equipment records are loaded from the database.
- Category and laboratory names are displayed using related data.
- This demonstrates the VIEW operation required for Presentation III.

Deletion of a Record

Demonstrate a real DELETE through the Equipment interface.

BEFORE DELETE

Select a test equipment record and click Delete.

DELETE

Confirm the deletion. The UI sends the DELETE operation

AFTER DELETE

Refresh the Equipment list and show that the record

DATABASE CHECK

Run `SELECT * FROM Equipment WHERE equipment_id = <deleted_id>`

Booking Interface

Booking records displayed using related user, equipment, laboratory and time-slot data.

DBMS PROJECT

Lab Equipment Booking & Maintenance

Dashboard

Equipment

Bookings

Consumables

Maintenance

Laboratory Equipment Booking & Maintenance System

Connected to MySQL: <http://localhost:4000>

Bookings

Bookings > Users > Equipment > Laboratory > Time_Slots

+ New Booking

BOOKING ID	USER	EQUIPMENT	LABORATORY	DATE	START	END	STATUS	ACTIONS
1001	Rahul	DSO-1000	Physics Lab	2026-08-27	10:00	11:00	Approved	Delete
1002	Priya	Digital Multimeter	Electronics Lab	2026-08-27	11:00	12:00	Pending	Delete
1003	Rahul	DSO-2000	Physics Lab	2026-08-28	09:00	10:00	Approved	Delete
1004	Priya	DSO-1000	Physics Lab	2026-08-29	10:00	11:00	Approved	Delete
1005	Priya	DSO-1000	Physics Lab	2026-08-27	10:00	11:00	Pending	Delete
1006	Ananya	Centrifuge C200	Chemistry Lab	2026-08-31	09:00	10:00	Approved	Delete
1007	Vikram	SolderPro X2	Electronics Lab	2026-08-31	10:00	11:00	Approved	Delete
1008	Neha	Function Generator FG1	Physics Lab	2026-08-31	11:00	12:00	Approved	Delete
1009	Ananya	DSO-1000	Physics Lab	2026-09-01	14:00	15:00	Pending	Delete
1010	Vikram	Digital Multimeter	Electronics Lab	2026-09-01	15:00	16:00	Approved	Delete
1011	Meera	Centrifuge C200	Chemistry Lab	2026-09-02	11:00	12:00	Approved	Delete
1012	Rohit	Function Generator FG1	Physics Lab	2026-09-02	14:00	15:00	Rejected	Delete

Supporting Modules

Additional UI screens demonstrate wider system functionality.

ID	NAME	CATEGORY	QUANTITY	MINIMUM STOCK
1	Electrodes	Physics Lab	50	10
2	Copper Wire	Electronics Lab	200	50
3	Water Filtration	Computer Lab	500	100
4	Laboratory Gloves	Chemistry Lab	5	20
5	Test Tubes	Chemistry Lab	100	50
6	Solder Wire	Electronics Lab	25	50
7	Printer Nozzles	Computer Lab	25	10
8	Connecting Wires	Physics Lab	150	30

Consumables – low-stock items highlighted

ID	EQUIPMENT	ISSUE/NOTE	VENDOR	START DATE	COMPLETION DATE	STATUS	REMARKS
1	Microwave X1	High: Microwave was malfunctioning	LabTech Services	2024-08-05	—	In Progress	Lines replacement in progress
2	SSD-3800	Medium: Diskley formatting	Precision Instruments	2024-08-20	2024-08-25	Completed	Display cable replaced
3	Centrifuge C200	High: Centrifuge motor failure	LabTech Services	2024-08-30	—	In Progress	Motor replacement underway
4	Substrate X2	Medium: Stocking to be confirmed	Precision Instruments	2024-09-01	2024-09-05	Completed	Substrate to be replaced
5	Centrifuge C200	Low: Thermal chamber being questioned	LabTech Services	2024-09-01	2024-09-10	Completed	Thermal issue resolved

Maintenance – jobs, breakdowns and vendors

Live Demonstration Flow

The UI acts as the front end for the existing MySQL database.



Conclusion

Presentation III demonstrates the working user interface.

- The interface provides live access to the laboratory equipment database.
- Equipment records can be viewed through the UI.
- New records can be inserted through the interface.
- Records can be deleted through the interface.
- Database changes can be verified directly in MySQL.
- The UI provides a simple front end for the Laboratory Equipment Booking & Maintenance System.

VIEW → INSERT → DELETE → VERIFY IN MYSQL